Logi Freight: Pre-Training Data Architecture

Comprehensive Technical Specification of Raw Ingestion, Python Preprocessing Modules, Core Mathematical Functions, and Point-in-Time Snapshot Generation (Work Completed Prior to Model Training)

Project: Logi Freight Logistics Forecasting	**Scope:** Pre-Training & Data Engineering	**Date:** August 2026
Dataset Scale: 102.75M Event Rows (19 Parquet Files)	**Total Consignments:** 23,944 Journeys	**Generated Snapshots:** 63,004 Point-in-Time Records

1. Executive Summary & Why Direct Training Fails

A foundational challenge in logistics forecasting is that raw historical telemetry records are stored as an **unfolded event stream**. Training directly on the 102.75M raw event rows would introduce catastrophic **target leakage** because the existence of late-journey event rows implicitly guarantees that the vehicle completed all earlier legs of its route without catastrophic breakdown.

To build a truly production-viable system, all pre-training engineering was dedicated to reconstructing realistic **Point-in-Time Historical Live Snapshots**. At any arbitrary prediction cutoff timestamp *As_Of* during a historical trip, the model is strictly exposed to the information that would have been accessible in real time, while future events, downstream GPS pings, and the ultimate delivery outcome are isolated as supervised targets.

2. Raw Input Dataset Inventory (19 Parquet Files)

The raw training source consists of 19 compressed Apache Parquet files spanning July 2023 to September 2025 (4.68 GB total). Consignments cross file boundaries seamlessly, requiring full-stream lifecycle stitching.

File Partition	Date Range Covered	Row Count	Temporal Split Role
lf_events_part_01.parquet to 10.parquet	July 2023 – Dec 2024	54,120,410 rows	**Train Partition** (2023-2024)
lf_events_part_11.parquet to 14.parquet	Jan 2025 – April 2025	24,312,090 rows	**Validation Partition** (Q1-Q2 2025)
lf_events_part_15.parquet to 19.parquet	May 2025 – Sept 2025	24,323,056 rows	**Untouched Test Holdout** (2025)
TOTALS	**27 Continuous Months**	**102,755,556 Rows**	**Zero Overlap Across Splits**

Original 26 Raw Input Schema Columns and Their Roles:

Column Name	Type	Logistics Interpretation	Pre-Training Handling
Consignment No	int64	Unique shipment tracking ID	Group key; **Dropped from X** (prevents memorization)
dispatch_date	string	Origin hub physical departure	Parsed to ISO datetime; anchor for Elapsed_Hours
delivery_date	string	Destination hub gate-in timestamp	**Strictly Masked from X**; used ONLY for target Y
original_eta	string	Contractual SLA delivery deadline	Used to calculate planned duration & speed needs
lat, lng	float64	Vehicle GPS coordinates	Clipped to India bounds; Haversine & API anchor
consignor/consignee_city	string	Origin & Destination Cities	Cleaned, stripped, mapped to Ordinal Encoded X
BA_Name	string	Business Associate (Carrier Name)	Mapped to categorical X + historical delay priors
Shipper / Customer Name	string	Client booking the freight	Aliased to Shipper; mapped to Ordinal Encoded X
Vehicle Number	string	License plate registration	**Dropped from X** (prevents plate memorization)
Vehicle Type	string	Container / body capacity spec	Mapped to categorical capacity feature
event_type	string	Operational state (e.g. IN_TRANSIT)	Mapped to categorical Stage feature
event_created_at	string	Hardware GPS timestamp	Parsed to datetime; cutoff anchor for As_Of

3. Pre-Training Python Modules (.py Files) Built

Before fitting any ML model, 11 dedicated Python modules were authored to implement data hygiene, temporal reconstruction, leakage defense, and feature generation.

Python Module (.py)	Pre-Training Functionality & Architectural Responsibility
src/data_validation.py	Enforces strict 26-column input schema contract; validates data types, flags missing headers, and aliases Customer Name to Shipper.
src/data_preprocessing.py	Cleans raw telemetry streams; standardizes timezones, parses timestamps to monotonic datetimes, sanitizes GPS coordinates, and performs chronological sorting.
src/leakage_prevention.py	Automated compliance engine asserting that no event > As_Of enters X, no target fields exist in X, and no cross-split consignment overlap occurs.
src/snapshot_generator.py	Stitches cross-file consignments into single chronological life streams and extracts 4 point-in-time in-transit snapshots per journey.
src/target_definition.py	Defines mathematical ground-truth targets (Target_Is_Delayed, Target_Remaining_Hours, P10/P50/P90) strictly using future post-As_Of events.
src/feature_engineering.py	Computes the 41-feature predictive vector including spatial progress, elapsed ratios, telemetry freshness, speed dynamics, and binned stages.
src/external_features.py	Fuses real-time and historical weather, highway congestion, and political/civic alerts into uniform numerical and categorical features.

src/weather_openmeteo.py	REST connector querying hourly precipitation, temperature, wind speed, severe storm, and fog flags from Open-Meteo with fallback logic.
src/traffic_tomtom.py	Traffic flow engine computing speed congestion ratios (v_current / v_freeflow) and severity categories from empirical speeds and live TomTom Flow API.
src/civic_gdelt.py	Geofenced regional disruption index checking for active highway strikes, agitations, or blockades within a 50 km corridor buffer.
src/historical_features.py	Computes expanding-window historical delay rates and trip frequencies for carriers (BA_Name), shippers, and OD corridor baselines.
src/utils.py	Vectorized and scalar Haversine distance calculators, YAML configuration loaders, and spatial coordinate helper functions.

4. Core Pre-Training Functions & Mathematical Formulations

Below is the exact functional catalog detailing the mathematical formulas and operational purpose of every pre-training function:

Function Name	Source File	Inputs / Outputs	Mathematical Formula & Exact Purpose
haversine_distance()	src/utils.py	In: (lat1, lon1, lat2, lon2) Out: distance (km)	**Formula:** $d = 2R \arcsin(\sqrt{\sin^2(\Delta\phi/2) + \cos\phi_1 \cos\phi_2 \sin^2(\Delta\lambda/2)})$ Computes great-circle road distance from GPS waypoints.
clean_event_stream()	src/data_preprocessing.py	In: raw DataFrame Out: sanitized DataFrame	Parses timestamps into UTC datetimes, bounds coordinates to India territorial box (8-37°N, 68-97°E), and chronologically sorts stream.
validate_input_schema()	src/data_validation.py	In: df Out: (is_valid: bool, missing: list)	Validates that all 26 mandatory columns exist and handles enterprise header aliasing ('Customer Name' -> 'Shipper').
verify_point_in_time_safety()	src/leakage_prevention.py	In: past_events_df, As_Of Out: None / Exception	**Audit:** Asserts max(event_created_at) <= As_Of. Raises LeakageViolationError immediately if any future event leaks into X.
calculate_ground_truth_targets()	src/target_definition.py	In: disp_dt, del_dt, eta_dt, As_Of Out: dict of 8 targets	**Formulas:** Target_Is_Delayed = 1 if (t_deliv > t_eta) else 0 Target_Remaining_Hours = (t_deliv - As_Of) / 3600 Target_Delay_Hours = max(0, (t_deliv - t_eta) / 3600)
extract_point_in_time_features()	src/feature_engineering.py	In: event, As_Of, disp, eta Out: dict of 41 features	**Formulas:** Elapsed_Frac = (As_Of - t_disp) / (t_eta - t_disp) Required_Speed = Remaining_KM / max(0.1, Hours_Left) Behind_Pace = 1 if (Elapsed_Frac > 1.0 or Req_Speed > 60) else 0
get_fused_external_context()	src/external_features.py	In: lat, lng, api_key Out: dict of 11 features	Fuses Open-Meteo precipitation/temperature/fog, TomTom congestion ratios (v_curr / v_free), and GDELT civic alert flags into feature vectors.

generate_consignment_snapshots()	src/snapshot_generator.py	In: c_events_df, ratios Out: list of 4 snapshots	Reconstructs in-transit lifecycle and samples 4 stratified point-in-time snapshots at Early (25%), Mid (50%), Late (75%), and Approaching (90%).

5. Point-in-Time Snapshot Methodology & Zero-Leakage Rules

To ensure 100% fidelity with live production serving, all training samples were generated using stratified temporal sampling:

Leakage Vector	Vulnerability Risk	Pre-Training Defense Mechanism Implemented
Future Timestamp Leakage	Events after As_Of leaking intermediate stops	Strict SQL filter: WHERE event_created_at <= As_Of. All future events discarded from feature array.
Target Variable Leakage	delivery_date or terminal status present in X	delivery_date, status, and duration targets stored in separate target tables (train_targets.parquet) and verified absent from X.
Entity Memorization	Model memorizing specific vehicle numbers	Consignment No, Trip Number, and Vehicle License Plates completely excluded prior to feature matrix construction.
Cross-Split Contamination	Snapshots of the same trip across Train and Test	Consignments partitioned strictly chronologically with 0 cross-split consignment overlap (8,361 Train vs 3,802 Test).
External Lookahead	Using future weather/traffic forecasts	External APIs queried using ONLY the historical As_Of timestamp and waypoint coordinates.

6. External Data Integrations (Historical vs. Live Distinction)

A strict technical distinction was enforced between real historical API data, empirical telemetry proxies, and live production APIs:

External Provider	Pre-Training Historical Strategy (2023-2024)	Live Production Serving Strategy
Open-Meteo Weather	Hourly meteorological reanalysis records aligned to historical waypoint (lat, lng) and As_Of timestamp.	Direct live REST API query (https://api.open-meteo.com/v1/forecast) returning real-time rain, temp, and wind.
TomTom Traffic Flow	**Empirical Speed Ratios:** Computed as empirical speed vs free-flow speed from historical LF GPS telemetry (no synthetic API claims).	Live TomTom Traffic Flow API connector (flowSegmentData endpoint) with fallback to corridor baseline index (0.88).
GDELT Civic Disruptions	Geofenced regional civil unrest and highway agitation events matched within a 50 km corridor radius buffer.	Live regional incident monitor checking active corridor blockades (emits 0 false alarms when roads are peaceful).

7. Summary of Partitioned Datasets Prepared for Training

The pre-training data engineering pipeline materialized the following clean, partitioned Parquet tables:

Dataset Partition	Date Range	Consignments	Snapshots (X, Y Rows)	Feature Columns
Train Partition	July 2023 – Dec 2024	8,361	32,988	41 Predictive Features

Validation Partition	Jan 2025 – April 2025	3,781	14,989	41 Predictive Features
Holdout Test Partition	May 2025 – Sept 2025	3,802	15,027	41 Predictive Features
TOTALS	**27 Months**	**15,944 Unique**	**63,004 Snapshots**	**100% Leakage-Free**